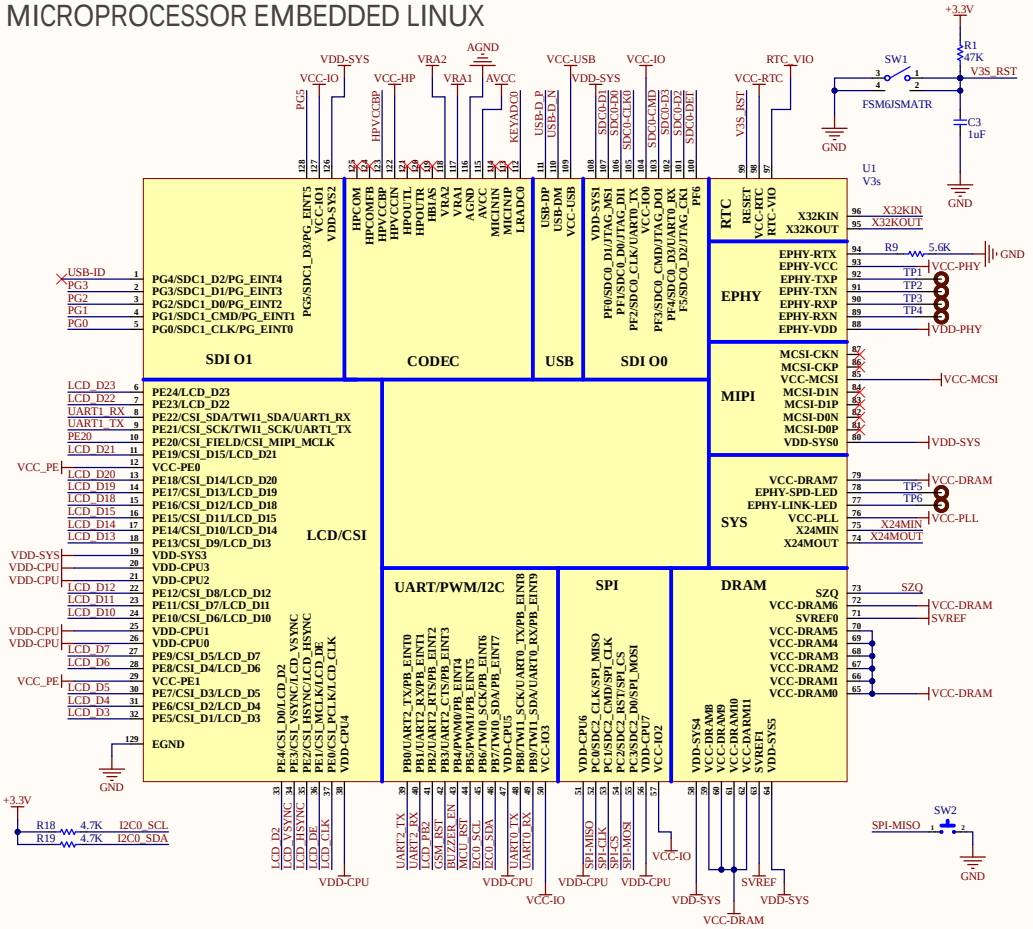
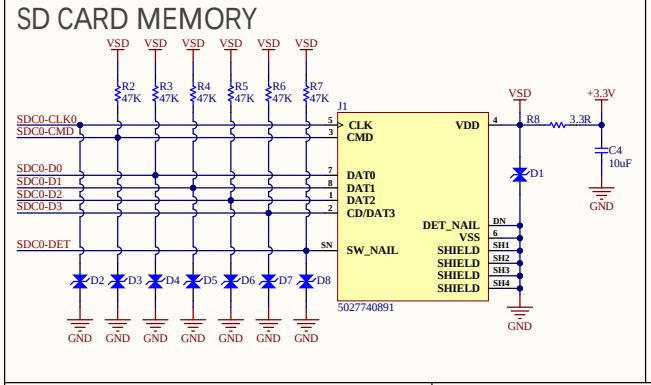
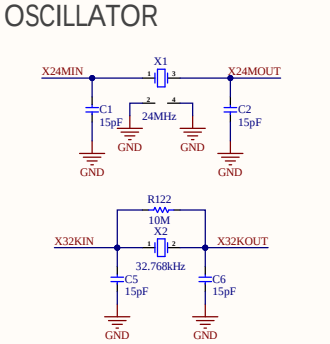
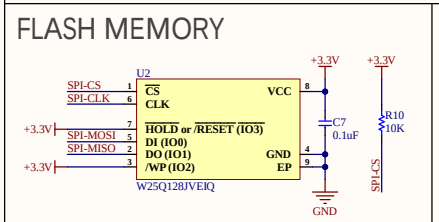
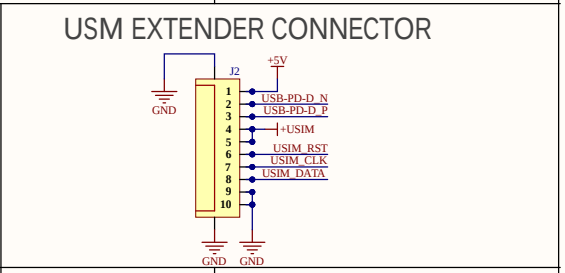
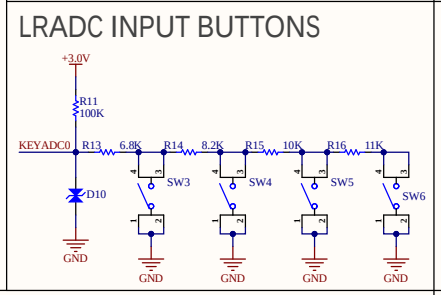


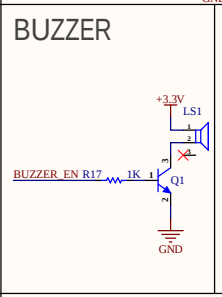


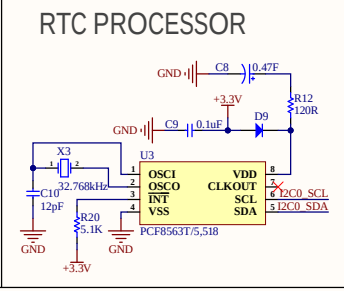


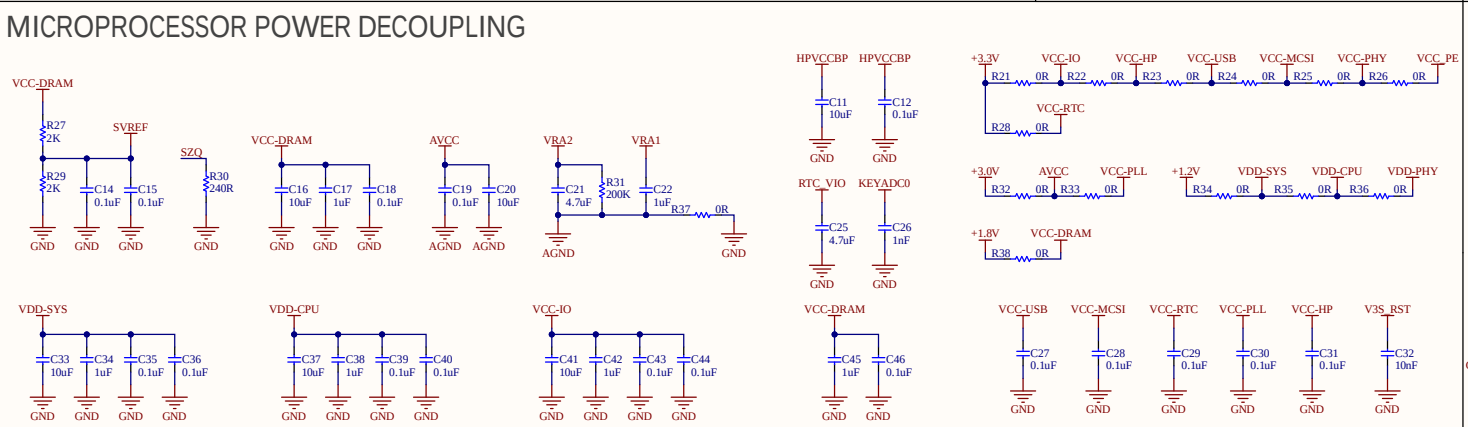


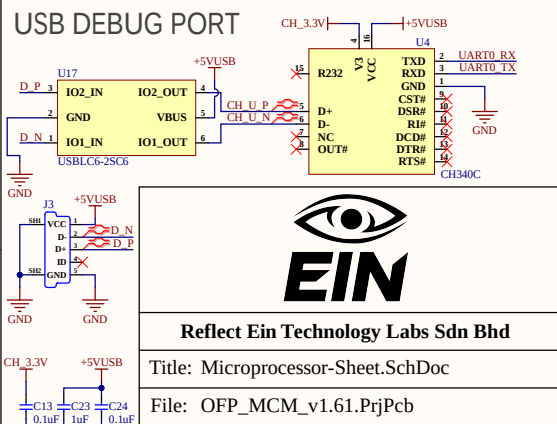












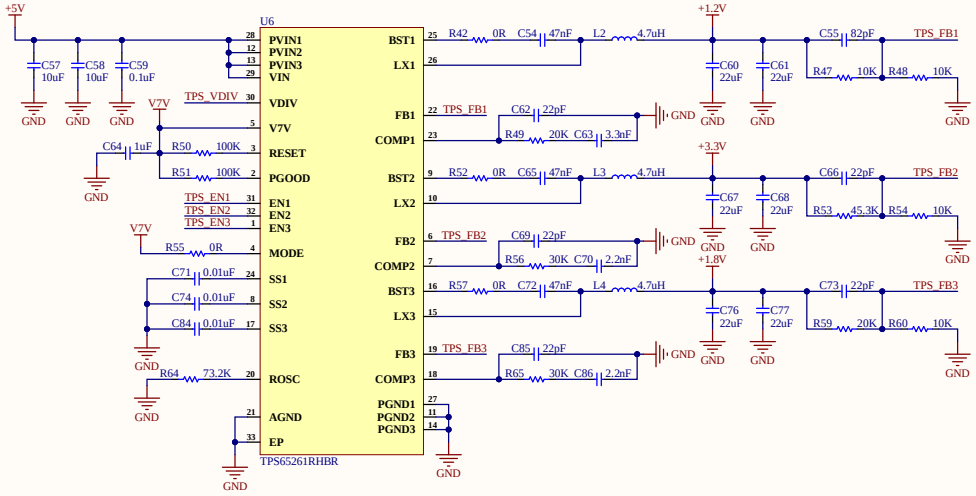
Reflect Ein Technology Labs Sdn Bhd

Title: Microprocessor-Sheet.SchDoc	
File: OFP_MCM_v1.61.PrjPcb	
Date: 9/26/2025	Page Num: 1
Drawn By: Nabil bin Feisal Bamadhaj	Page Total: 5

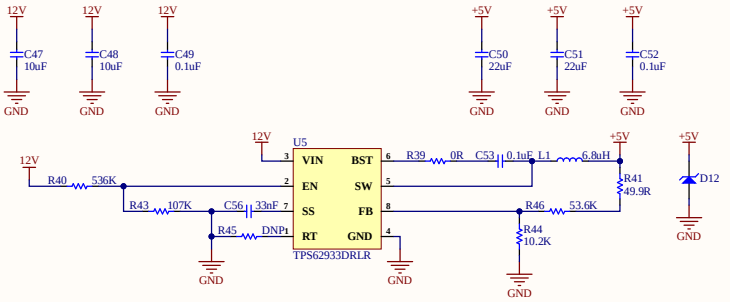
5V TO 3.3V BUCK CONVERTER

5V TO 1.8V BUCK CONVERTER

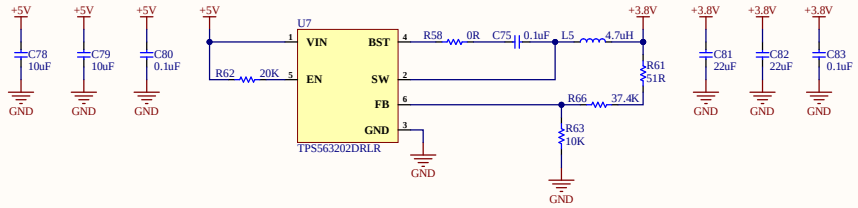
5V TO 1.2V BUCK CONVERTER



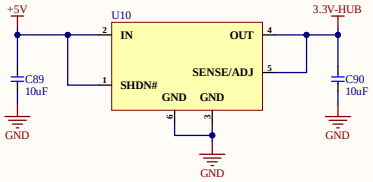
12V TO 5V BUCK CONVERTER (SYSTEM)



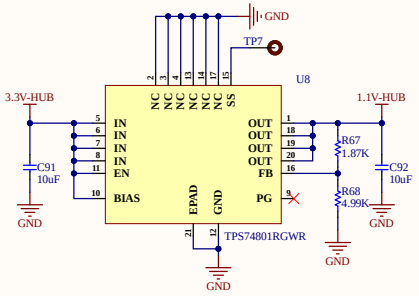
5V TO 3.8V BUCK CONVERTER (GSM)



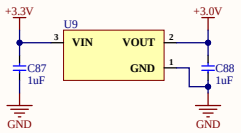
5V TO 3.3V LDO USB HUB



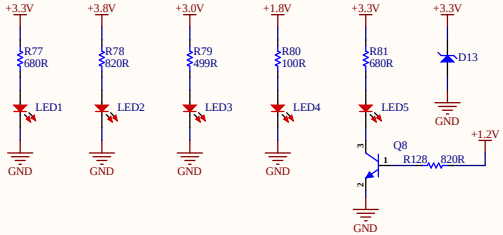
3.3V TO 1.1V LDO USB HUB

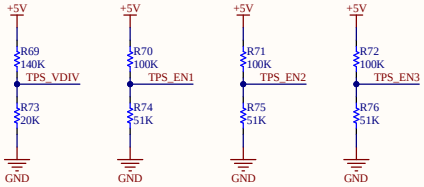


3.3V TO 3.0V LDO (MPU)



POWER INDICATORS

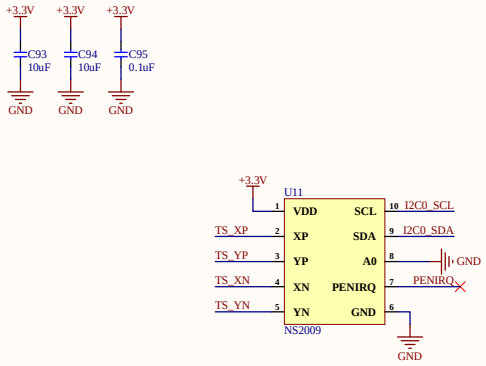




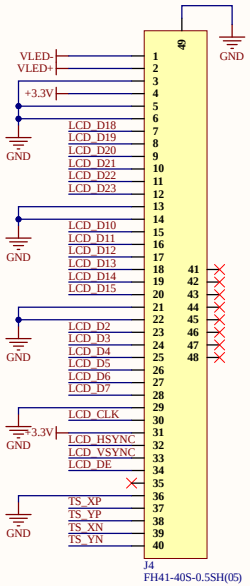
Reflect Ein Technology Labs Sdn Bhd

Title: Power-Sheet.SchDoc	
File: OFP_MCM_v1.61.PrjPcb	
Date: 9/26/2025	Page Num: 2
Drawn By: Nabil bin Feisal Bamadhaj	Page Total: 5

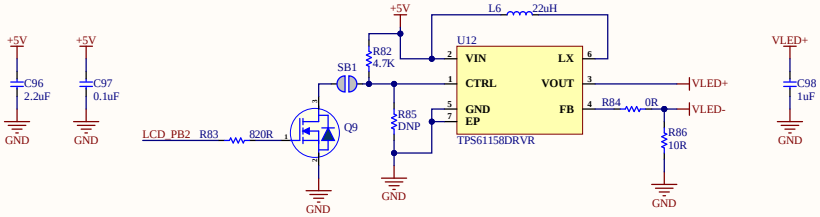
RGB LCD RESISTIVE TOUCH CONTROLLER



RGB LCD FPC/FFC CONNECTOR



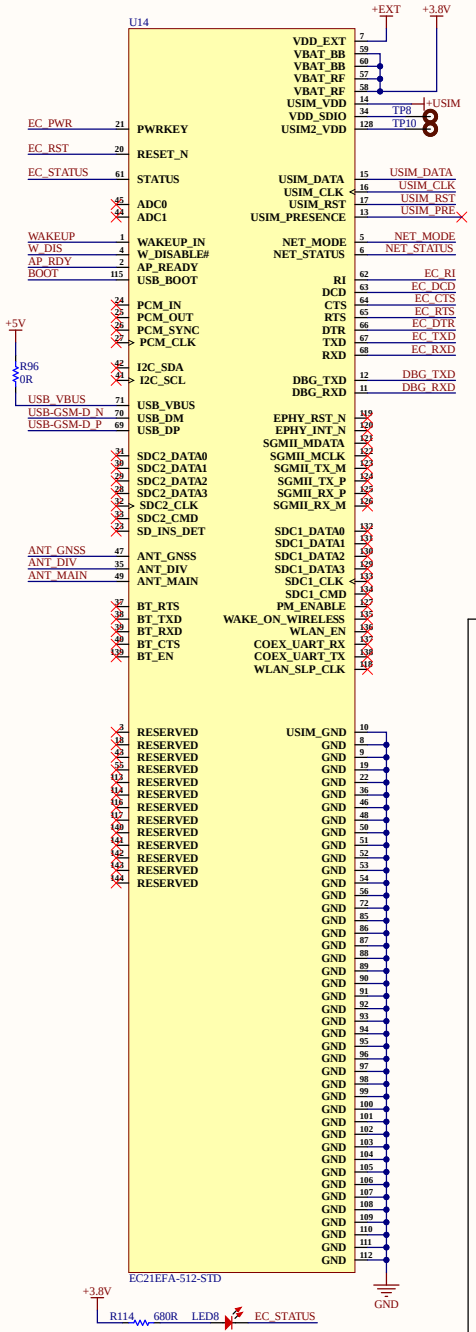
RGB LCD POWER CONTROLLER

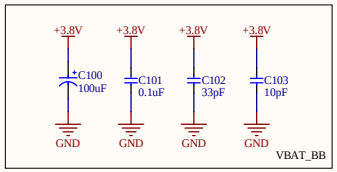


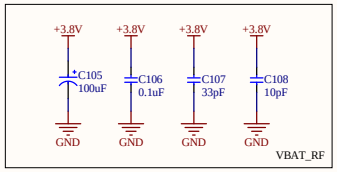
Reflect Ein Technology Labs Sdn Bhd

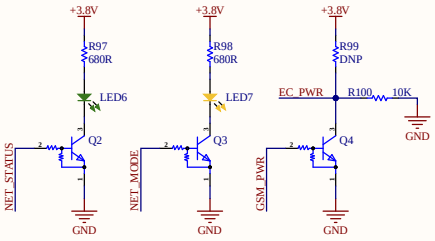
Title: RGB-LCD-Sheet.SchDoc	
File: OFP_MCM_v1.61.PrjPcb	
Date: 9/26/2025	Page Num: 3
Drawn By: Nabil bin Feisal Bamadhaj	Page Total: 5

EC21 GSM STANDARD MODULE

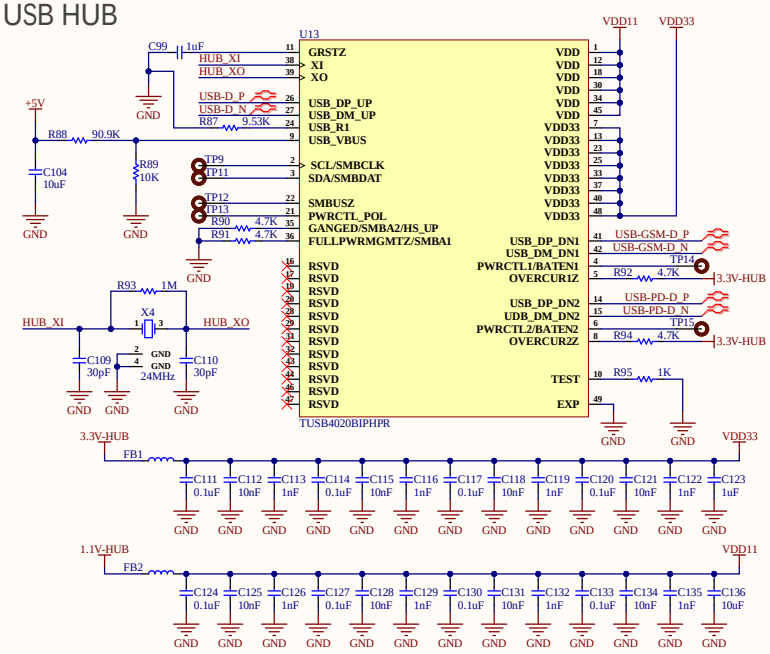




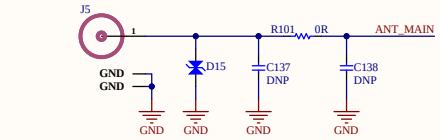


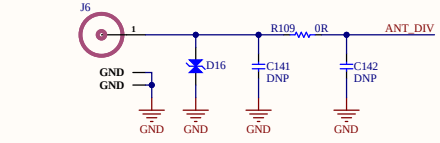


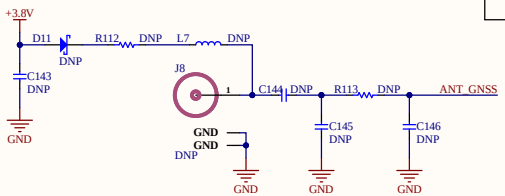
GSM USB HUB



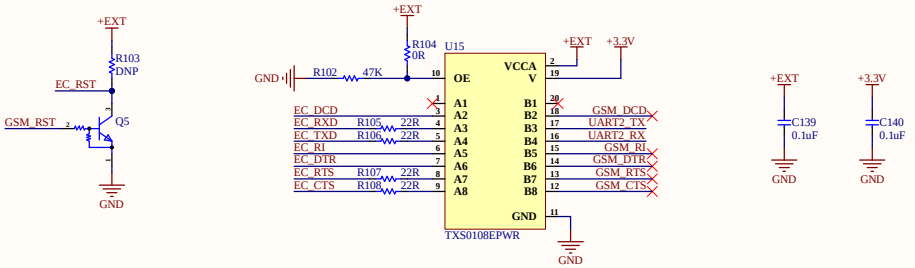
GSM ANTENNA CONNECTORS



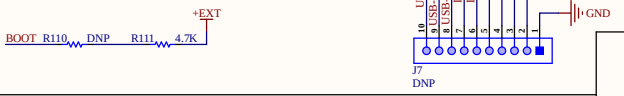




GSM 1.8V TO 3.3V LEVEL SHIFTER



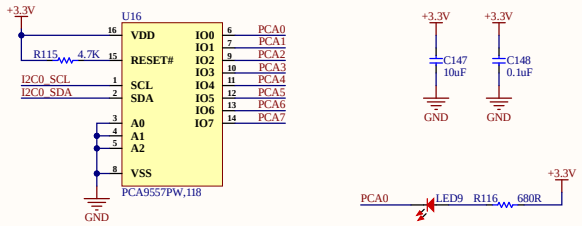
GSM OTHER UNUSED PINS



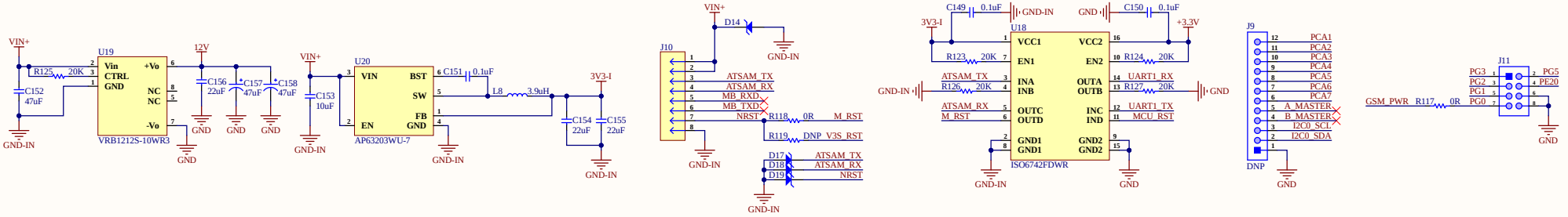
Reflect Ein Technology Labs Sdn Bhd

Title: GSM-USB-HUB-Sheet.SchDoc		
File: OFP_MCM_v1.61.PrjPcb		
Date: 9/26/2025	Page	Num: 4
Drawn By: Nabil bin Feisal Bamadhaj	Page	Total: 5

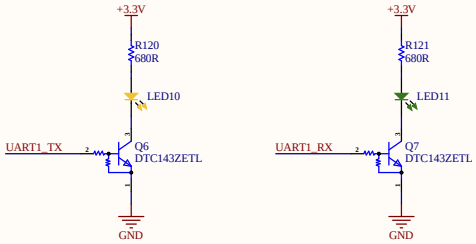
IO EXPANDER



CONNECTORS



TX / RX LED INDICATOR



Reflect Ein Technology Labs Sdn Bhd

Title: Others-Sheet.SchDoc	
File: OFP_MCM_v1.61.PrjPcb	
Date: 9/26/2025	Page Num: 5
Drawn By: Nabil bin Feisal Bamadhaj	Page Total: 5

Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 2	PP-006	0.203mm	4.4	
2	Layer 1	CF-004	0.035mm		
	Dielectric 1	FR-4	1.030mm	4.6	
3	Layer 2	CF-004	0.035mm		
	Dielectric 3	PP-006	0.203mm	4.4	
4	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.010mm	3.5	
	Bottom Overlay				
Total board thickness:			1.596mm		

